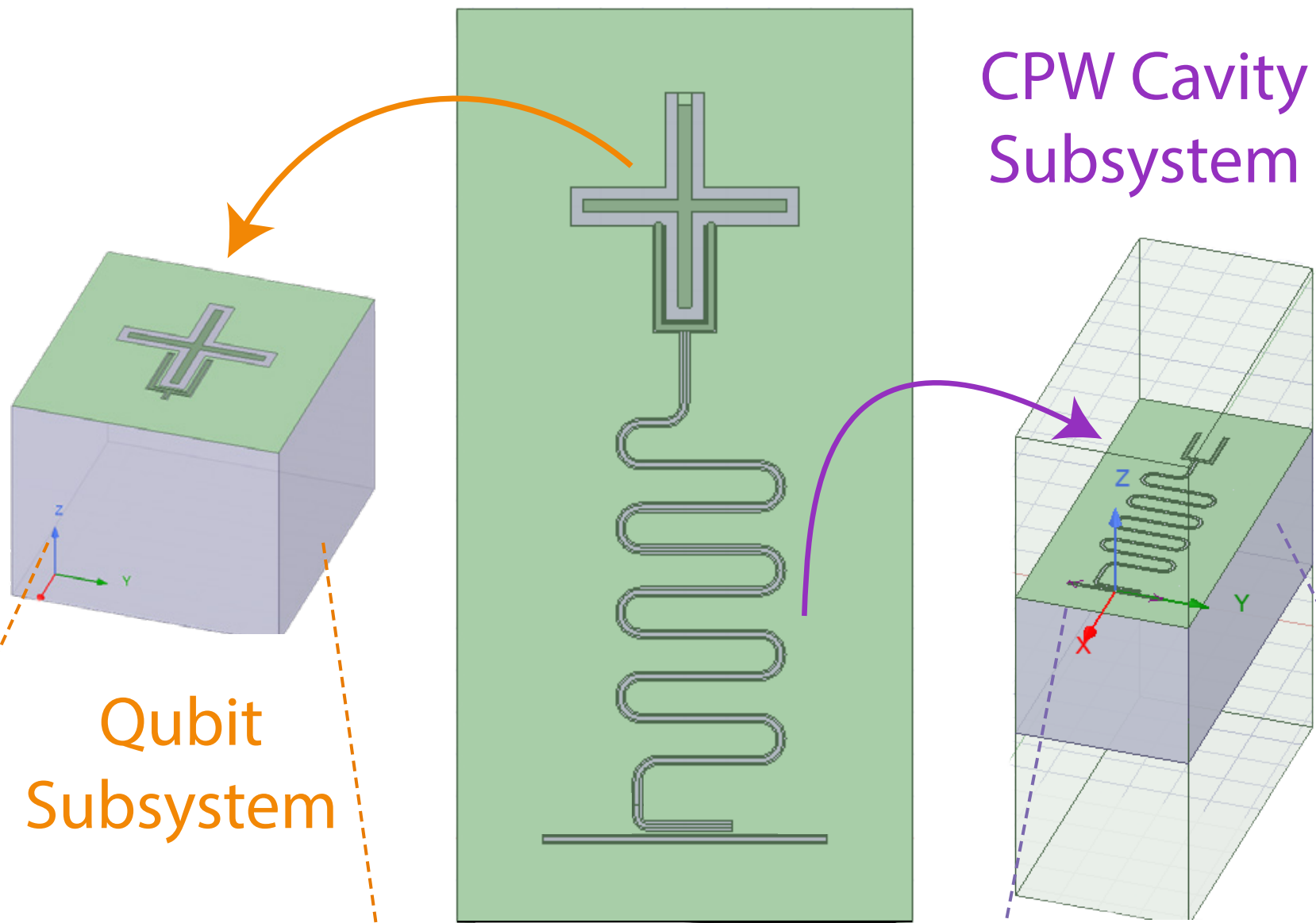




CPW Cavity
Subsystem

Qubit
Subsystem

α (anharmonicity)

$\approx -E_C$ from Q3D

f_{qubit} (qubit frequency)

$\approx \sqrt{(8 E_J E_C) - E_C}$ from Q3D

f_{res} (resonator freq.)

HFSS eigenmode solver

κ (resonator linewidth)

HFSS via 50 Ω port loss